ENVS 193DS Final

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Link to GitHub repository

Set Up

Created repository, added description and README

```
# read in packages
library(tidyverse)
library(here)
library(janitor)
library(DHARMa)
library(ggeffects)
library(MuMIn)
library(scales)
library(rstatix)

# read in nest observation data

nests <- read_csv(here("data", "nest_data_final.csv"))

# read in personal dinner cooking data
cooking <- read_csv(here("data", "193ds_personal_project.csv")) |>
  clean_names()
```

Problem 1. Research writing

a. Transparent statistical methods

In part 1, they used a binomial generalized linear model (aka logistic). The response is a binary outcome: the probability of American Avocet microhabitat use (it either is used or it is not). The predictor is distance to water edge, so it is continuous. This makes sense to use a binomial GLM to test whether a continuous predictor variable relates to a binary response.

In part 2, they used a chi-square test. Both the response (habitat type, 5 levels) and the predictor (bird species, 3 levels) variables are categorical. It makes sense to use chi-square because they're seeing if a relationship exists between categories.

b. Suggestions for rewriting

Part 1: The probability that American Avocet's use microhabitat depends on how far the microhabitat is from the water's edge

Part 2: There is a difference in the habitats that the three bird species (American Avocet, Forster's Tern, and Black-necked Stilt) use, so there are some species that are found in certain habitats more than others.

c. Further analysis

An additional variable that might influence the probability of American Avocet microhabitat use is the **water depth**. This is a continuous variable, measured in centimeters with a marked pole. As with most wading birds, American Avocets primarily feed by sweeping their bills through the water. It is thus likely that a microhabitat with shallower water will be used more than one with deep water, as it is easier to feed in.

Problem 2. Data analysis

a. Clean and wrangle

```
# create new object using nest data frame
nests_clean <- nests |>
  # keep only the Crematogaster ant species
  filter(ant_species %in% c("AB", "RRB", "BBR")) |>
  # create new column with the full scientific name
  mutate(scientific_name = case_match(
    ant_species,
    "AB" ~ "Crematogaster sjostedti",
    "RRB" ~ "Crematogaster mimosae",
    "BBR" ~ "Crematogaster nigriceps"
  )) |>
  # set the level order
  mutate(scientific_name = as_factor(scientific_name),
         scientific_name = fct_relevel(scientific_name,
                                       "Crematogaster sjostedti",
                                       "Crematogaster mimosae",
                                       "Crematogaster nigriceps")) |>
  # keep the three relevant columns
  select(case_control, height_cm, scientific_name)
```

```r
# display the structure of the cleaned data frame
str(nests_clean)
```

```
tibble [270 x 3] (S3: tbl_df/tbl/data.frame)
 $ case_control   : num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ height_cm      : num [1:270] 392 310 513 154 324 ...
 $ scientific_name: Factor w/ 3 levels "Crematogaster sjostedti",..: 2 2 2 2 1 2 3 2 1 3 ...
```

```r
# show 10 random rows
slice_sample(nests_clean, n = 10)
```

```
# A tibble: 10 x 3
   case_control height_cm scientific_name
          <dbl>     <dbl> <fct>
 1            1      288. Crematogaster mimosae
 2            0      598  Crematogaster mimosae
 3            0      249  Crematogaster mimosae
 4            0      144  Crematogaster mimosae
 5            0      212. Crematogaster mimosae
 6            0      123  Crematogaster mimosae
 7            0      170  Crematogaster nigriceps
 8            1      212. Crematogaster mimosae
 9            0      300  Crematogaster mimosae
10            0      213  Crematogaster mimosae
```

b. Response variable

The 1s and 0s in the "case_control" column tell if a tree had a bird nest or not. A 1 means the tree was observed to have one and the 0 meant the tree was an available tree without a nest.

c. Purpose of study

Tree height is a continuous variable measured in centimeters. As the tree height increases, the probability of a nest occuring should increase too, because it physically places a nest farther up from the ground and its predators. The paper did find that the probability of a bird nesting in a tree increased with height, as described in the *Results* section, *second* paragraph.

Acacia ant species is a categorical variable with three levels (C. sjostedti, C. mimosae, and C. nigriceps). It makes sense that the probability of nest occurring should depend on the ant species present in a tree. Trees that are defended by more aggressive ant species are more likely to have a nest because of the protection it will receive. The paper highlights this in the *Discussion* section, *first* paragraph.

d. Table of models

Model number	Model description
1	tree height and ant species as predictors, with an interaction term
2	tree height and ant species as predictors, without an interaction term

e. Running the models

```
# model 1: effect of tree height and ant species on nest presence
# w/ interaction term
model1 <- glm(
  # formula: response ~ variable * variable
  case_control ~ height_cm * scientific_name,
  # use clean data frame
  data = nests_clean,
  # binomial family for response
  family = "binomial"
)

# model 2: effect of tree height and ant species on nest presence
# w/o interaction term
model2 <- glm(
  # formula: response ~ variable + variable
  case_control ~ height_cm + scientific_name,
  # use clean data frame
  data = nests_clean,
  # binomial family for response
  family = "binomial"
)
```

f. Selecting the best model

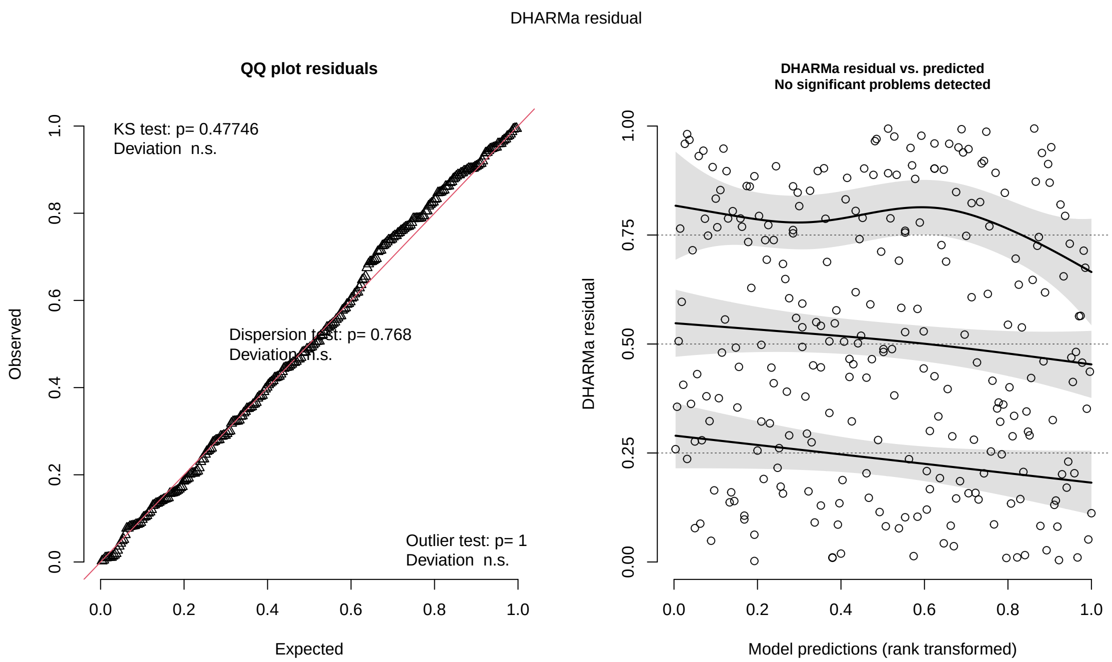
```r
# compare models using AICc
AICc(model1,
     model2) |>
  arrange(AICc)
```

```
       df     AICc
model1  6 238.1236
model2  4 258.8940
```

The best model, according to Akaike's Information Criterion (AIC) uses tree height and acacia ant species as predictors of the probability of bird nest occurrence *with* an interaction.

g. Checking the diagnostics

```r
# simulate residuals and plot diagnostics
plot(simulateResiduals(model1))
```

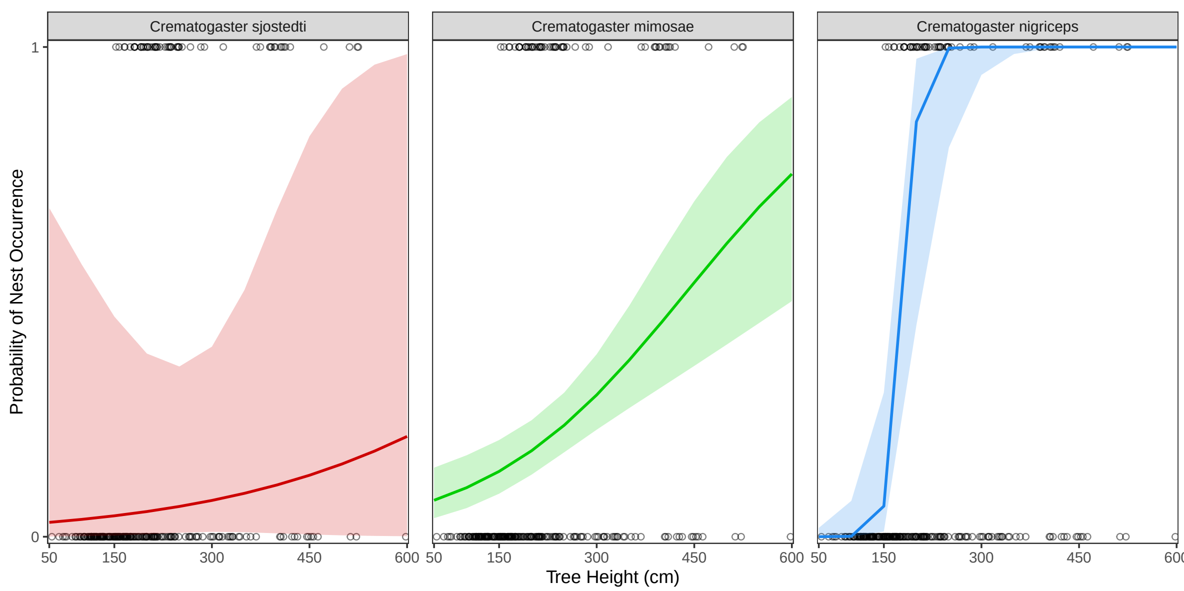


h. Visualizing model predictions

```
# generate model predictions for height across each ant species
model_preds <- ggpredict(
  # selected model
  model1,
  # predictors
  terms = c("height_cm", "scientific_name")
)

# base layer: ggplot
ggplot(data = nests_clean,
       # tree height x-axis
       # nest occurence on y-axis
       mapping = aes(x = height_cm,
                     y = case_control)) +
  # plotting raw nest points
  geom_point(shape = 21,
             alpha = 0.5,
             col = "black") +
  # adding 95% confidence interval for ant species
  geom_ribbon(data = model_preds,
              aes(x = x,
                  y = predicted,
                  ymin = conf.low,
                  ymax = conf.high,
                  fill = group),
              alpha = 0.2) +
  # adding prediction lines for each ant
  geom_line(data = model_preds,
            aes(x = x,
                y = predicted,
                color = group),
            linewidth = 0.8) +
  # make y-axis to the 0-1 probability
  scale_y_continuous(limits = c(0, 1),
                     breaks = c(0, 1),
                     expand = c(0.007, 0.007)) +
  # make x-axis the range of tree heights
  scale_x_continuous(limits = c(50, 600),
                     breaks = c(50, 150, 300, 450, 600),
                     expand = c(0.005, 0.005)) +
```

```
  # non-default theme
  theme_bw() +
  theme(panel.grid = element_blank(),
        # hide redundant legend
        legend.position = "none",
        panel.spacing.x = unit(0.5, "cm")) +
  # labellinng axes
  labs(x = "Tree Height (cm)",
       y = "Probability of Nest Occurrence") +
  # assign line colors to each ant
  scale_color_manual(values = c("Crematogaster sjostedti" = "red3",
                               "Crematogaster mimosae" = "green3",
                               "Crematogaster nigriceps" = "dodgerblue2")) +
  # manually assigning ribbon colors to each ant
  scale_fill_manual(values = c("Crematogaster sjostedti" = "red3",
                              "Crematogaster mimosae" = "green3",
                              "Crematogaster nigriceps" = "dodgerblue2")) +
  # create separate panel for each ant
  facet_wrap(~ group)
```



i. Writing figure caption

Figure 1. The probability of bird nest occurrence increases with tree height and the strength of the relationships differs amongst acacia ant species type. Open black circles represent individual

trees, plotted by height (cm) and whether they had a bird nest (1) or no nest (0). Colored lines represent predicited probability of nest occurence from a binomial generalized linear model *with* an interaction between tree height and ant species. Shaded bands show the 95% confidence intervals about these prediction lines. Beyond colors, the separate panels distinguish each of the three ant species: Crematogaster sjostedti (red), Crematogaster mimosae (green), and Crematogaster nigriceps (blue).

Data is sourced from Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. https://doi.org/10.5281/zenodo.8373322

j. Calculating model predictions

```
# predict nest probability at the tallest tree (600 cm) for each ant species
ggpredict(model1, terms = c("height_cm [600]", "scientific_name"))
```

```
# Predicted probabilities of case_control

scientific_name: Crematogaster sjostedti

height_cm | Predicted |     95% CI
-----------------------------------
      600 |      0.20 | 0.00, 0.99

scientific_name: Crematogaster mimosae

height_cm | Predicted |     95% CI
-----------------------------------
      600 |      0.74 | 0.48, 0.90

scientific_name: Crematogaster nigriceps

height_cm | Predicted |     95% CI
-----------------------------------
      600 |      1.00 | 1.00, 1.00
```

k. Interpreting results

At the tallest tree height (600 cm), the predicted probability of nest occurence is 0.20 (95% CI: [0.00, 0.99]) for Crematogaster sjostedti, is 0.74 (95% CI: [0.48, 0.90]) for Crematogaster mimosae, and is 1.00 (95% CI: [1.00, 1.00]) for Crematogaster nigriceps (Figure 1). As tree height increases, the probability of nest occurrence generally increases, though the strength of this relationship depends on the ant species. It is strongest for Crematogaster nigriceps, intermediate for Crematogaster mimosae, and weakest for Crematogaster sjostedti. This means birds are most likely to nest in tall trees with Crematogaster nigriceps and Crematogaster mimosae, and rarely nest in trees with Crematogaster sjostedti. Biologically, this shows the protection from predators: Crematogaster mimosae and Crematogaster nigriceps will aggressively defend host trees, so birds benefit from nesting in these trees.

Problem 3. Affective and exploratory visualizations

a. Comparing visualizations

- The exploratory plots from Homework 2 represents the dinner cook time literally: each dinner is a point sitting at its exact day on the x-axis, with the means marked so the numbers can be read directly. My affective visualization represents just the average cook times, while encoding the length and intensity of each cook through the height and density of the meal illustrations laid out in a weekly planner. Exploratory was all data, but the other has more emotion and feeling.
- All of my visualizations use the same dinner cooking dataset, and all of them organized the cook time around the structure of the week, making the similar patterns show up even more. The visuals that had cook time on the vertical/y-axis specifically made me think about adjusting the height of a stacked meal in a planner column for affective visualization.
- The visualizations show the same patterns: cook times tend to be longer on weekends and on more involved days. They are lighter on rushed weekdays, but nights with a helper also tended to run longer no matter the day. The affective piece simply communicates that pattern through density of objects and height rather than their exact values, so it reads as a rhythm instead of just some measurement.
- In Workshop 9 I got feedback that my original hand-drawn plan would be hard to finish cleanly in the time I had, and that a digital format would let the audience understand more. I took that suggestion and rebuilt the planner in Canva, which let me keep the same idea (meal complexity and stack height standing in for cook time and stress) but I could have the columns more legible and have consistent styles.

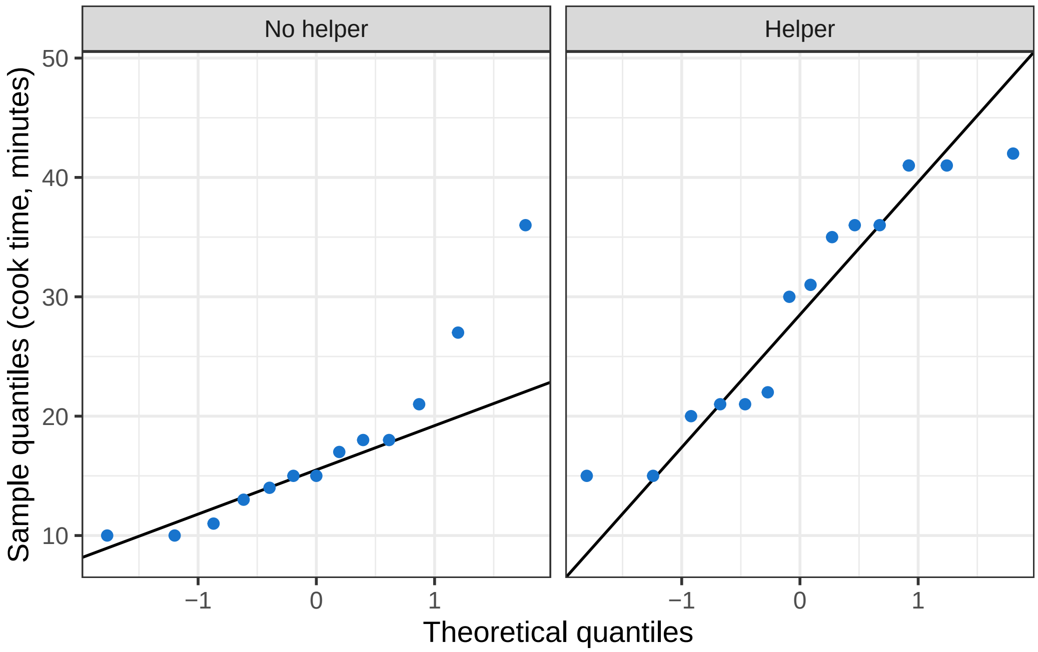
b. Designing an analysis

My response variable is dinner cook time, which is continuous and measured in minutes. The predictor I am using is helper presence, a categorical variable with two levels (no helper and helper). To answer whether having a helper changes how long dinner takes to cook, I will run a two-sample t-test comparing mean cook time between the two groups. A t-test is appropriate here because I am comparing the means of a continuous response across exactly two levels of a categorical predictor, which is the exact situation a two-sample t-test is built for.

c. Checking assumptions and running analysis

```r
# clean and prep cooking data for the analysis
cooking_clean <- cooking |>
  # keep only the columns I need
  select(cook_time, helpers) |>
  # drop rows missing either value
  drop_na() |>
  # label the two helper levels (only ever 0 or 1)
  mutate(helpers = factor(helpers,
                          levels = c(0, 1),
                          labels = c("No helper", "Helper")))

# normality: QQ plot of cook time within each helper group
ggplot(data = cooking_clean,
       mapping = aes(sample = cook_time)) +
  geom_qq_line(color = "black") +
  geom_qq(color = "dodgerblue3") +
  facet_wrap(~ helpers) +
  labs(x = "Theoretical quantiles",
       y = "Sample quantiles (cook time, minutes)") +
  theme_bw()
```




```
# equal variance: F-test comparing the two groups' variances
var.test(cook_time ~ helpers, data = cooking_clean)
```

```
	F test to compare two variances

data:  cook_time by helpers
F = 0.56059, num df = 12, denom df = 13, p-value = 0.3251
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.1777871 1.8159113
sample estimates:
ratio of variances 
         0.5605939 
```

I checked normality visually with QQ plots for each group. For no helper, the points fall reasonably along the line with some deviation at the upper tail, so normality is accepted. For 1 helper, it is a little less tight, but more consistently tracking the line. I tested equal variances with an F-test ($F = 0.56$, $p = 0.33$). Because $p>0.05$ I fail to reject equal variances, so assumption was met. I will run Welch's two-sample t-test. Finally, observations were independent because each value is a separate dinner cooked on a different night.

```r
# Welch's two-sample t-test
t.test(cook_time ~ helpers,
       data = cooking_clean,
       var.equal = FALSE)
```

```

	Welch Two Sample t-test

data:  cook_time by helpers
t = -3.5287, df = 23.97, p-value = 0.001718
alternative hypothesis: true difference in means between group No helper and group Helper is
95 percent confidence interval:
 -18.531487  -4.853128
sample estimates:
mean in group No helper    mean in group Helper 
               17.30769                29.00000 
```

```r
# Cohen's d effect size
cohens_d(cooking_clean, cook_time ~ helpers)
```

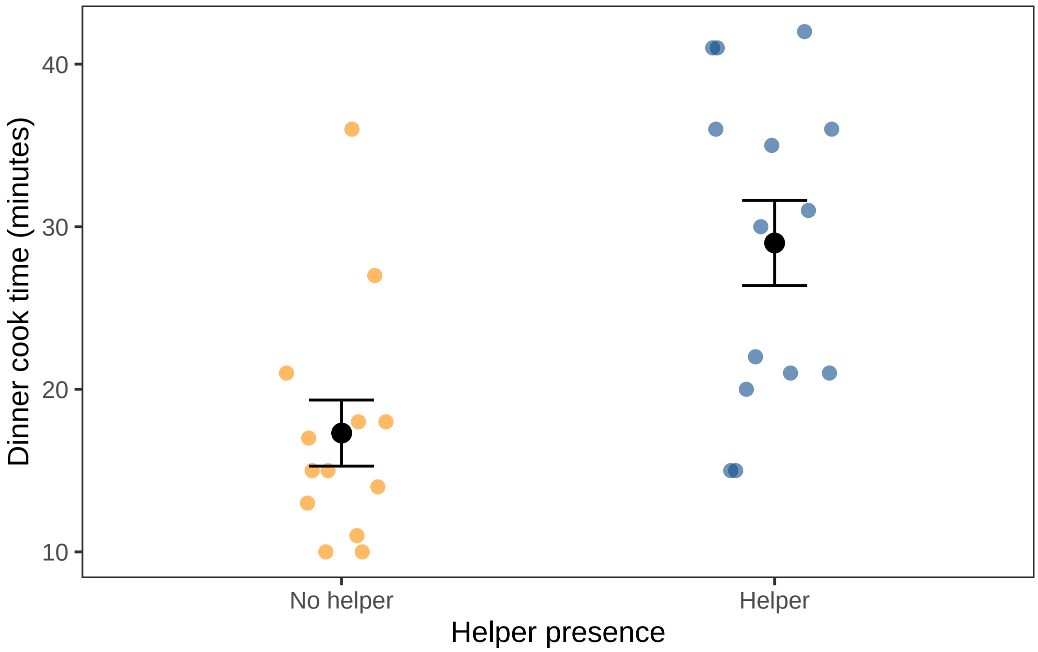
```
# A tibble: 1 x 7
  .y.       group1    group2 effsize    n1    n2 magnitude
* <chr>     <chr>     <chr>    <dbl> <int> <int> <ord>    
1 cook_time No helper Helper   -1.35    13    14 large    
```

d. Creating a visualization

```r
# group means and standard errors for the overlay
cooking_summary <- cooking_clean |>
  group_by(helpers) |>
  # get mean, then alculated se
  summarize(mean_time = mean(cook_time),
            se_time = sd(cook_time) / sqrt(n())) |>
  ungroup()

# base layer: cook time by helper presence
ggplot(data = cooking_clean,
       mapping = aes(x = helpers,
                     y = cook_time,
```

```
                    color = helpers)) +
# raw data
geom_jitter(width = 0.15, height = 0,
            alpha = 0.6, size = 2) +
# standard error bars around each mean
geom_errorbar(data = cooking_summary,
              inherit.aes = FALSE,
              aes(x = helpers,
                  ymin = mean_time - se_time,
                  ymax = mean_time + se_time),
              width = 0.15, color = "black") +
# group mean points
stat_summary(geom = "point", fun = mean,
             size = 3, color = "black") +
# custom colors
scale_color_manual(values = c("No helper" = "darkorange",
                              "Helper" = "dodgerblue4")) +
# label axes with units
labs(x = "Helper presence",
     y = "Dinner cook time (minutes)") +
# non-default theme, remove gridlines, no legend
theme_bw() +
theme(panel.grid = element_blank(),
      legend.position = "none")
```



e. Writing a caption

Figure 2. Dinner cook time (minutes) on nights cooked alone versus nights with a helper present. Colored points show individual dinners (orange = no helper, blue = a helper). Black points show the mean cook time for each group and black error bars show +/- 1 standard error from the mean.

f. Writing about results

I used a Welch's two-sample t-test to compare mean dinner cook time between nights cooked alone and nights with a helper present. Cook time was significantly higher with a helper (mean = 29.0 min) than when done alone (mean = 17.3 min) for dinner (Welch's $t(23.97) = -3.53$, $p = 0.0017$, $\alpha = 0.05$). The effect size was large (Cohen's $d = -1.35$), indicating that helper presence is associated with a meaningful increase in cook time.

g. Sharing affective visualization

Present in class.